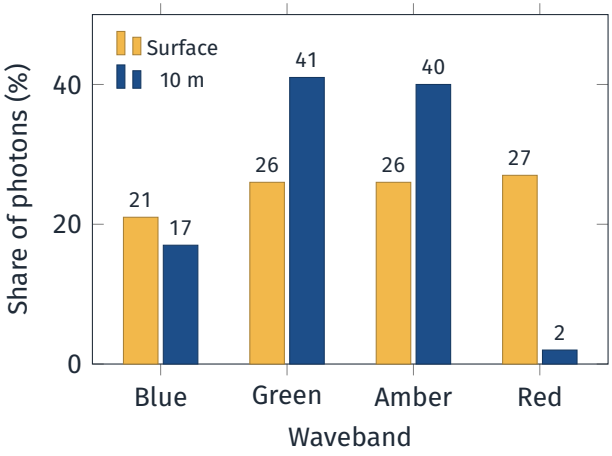
The light field at the study site

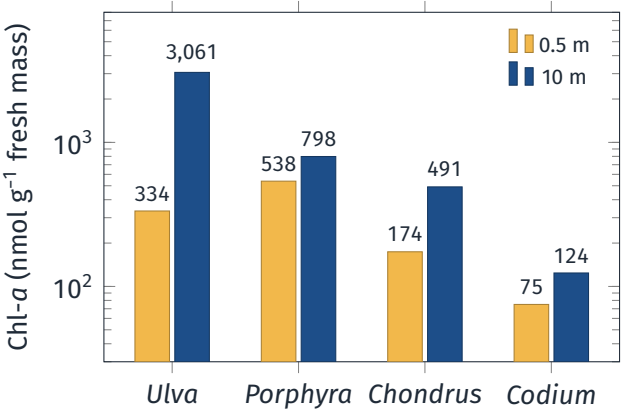
- Great Harbor, Woods Hole (USA), summer 1974–75
- Depth over which photon flux falls ten-fold:
  - red 3.6 m; blue 5.3 m
  - amber 6.0 m; green 6.0 m
- At 10 m: 1.45% of surface photons remain
- Coastal water turns the light **green-yellow**, not blue
- Depth reduces both **quantity** and **quality**



Bands: blue 430–510, green 510–560, amber 560–620, red 620–710 nm. Data: Ramus et al. (1976b), Table 1.

Pigment content after 7 days at 0.5 m or 10 m

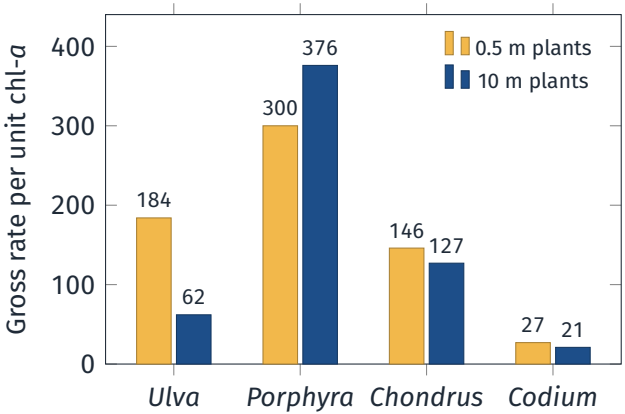
- All four species made **more pigment** at 10 m
- *Ulva*: about 9× more chl-*a*; chl-*b*:chl-*a* 0.47 → 0.61
- *Codium*: about 1.7× more chl-*a*; chl-*b*:chl-*a* unchanged (0.65, 0.68)
- Red algae: phycoerythrin:chl-*a* about doubled
- Changes reversed when lines were turned upside down



Log scale. Data: Ramus et al. (1976a), Figs 1–4 captions; reversal: Ramus et al. (1976b).

Light-saturated photosynthesis per unit chlorophyll-*a*

- Laboratory rates at saturating light
- *Ulva*: rate per chl-*a* fell to one third at 10 m
  - extra chlorophyll enlarges the antennae, not the capacity to use bright light
- Other species: changes within about 25%
- Deep *Ulva* and *Codium* saturated at half the irradiance



Units as in Ramus et al. (1976a), Table I; 26 °C, 10 mW cm^{-2}.

Three ways to harvest dim, green light

1. More pigment

Increase chl-*a* and accessory pigments per unit area

e.g. *Ulva* at 10 m: about 9× more chl-*a*

2. Shift the pigment ratio

More accessory pigment relative to chl-*a* (any accessory pigment, not only phycoerythrin)

e.g. phycoerythrin:chl-*a* doubled in *Chondrus*

3. Build an absorber

Thick, scattering thallus that traps light (optically black)

e.g. *Codium* absorbs 97–99% at any depth

None of these is unique to red algae; irradiance, not colour, drives most of the response

After Saffo (1987); data from Ramus et al. (1976a) and Ramus (1978).